DO NOW — RECALL

Lesson 101 · Multicore CPUs and Parallel Systems

Name: _____ Class: _____ Date: _____

1. What is a core?

2. Define concurrency.

3. What does Amdahl's law illustrate?

4. A 3×4 array of independent cells: how many cells?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A CPU processing unit able to run a hardware thread or more, depending on design.

2. Model answer

Several tasks progress over overlapping time.

3. Model answer

Serial work limits parallel speedup.

4. Model answer

12.

By the end, you can…

• A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

• Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

• Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

Vocabulary — know these terms

multicore — Processor with multiple CPU cores.

parallelism — Simultaneous processing of independent tasks.

critical path — Longest dependent chain.

Amdahl's law — Speedup bounded by serial portion.

NEW LEARNING

Read, annotate and explain

Understand

A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

Apply

Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

Evaluate

Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Eight equal independent jobs are shared across two identical cores. How many ideal waves?

Explain the choice

8 ÷ 2 = 4 equal-duration waves, versus 8 on one core.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Twelve equal independent jobs run on three identical cores. Compare ideal waves with one core. [6]

Question 2

Explain two benefits and two limitations of multicore processing for video rendering. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Half a program must run serially; other half parallelised perfectly across four cores. Compute ideal speedup and interpret. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Four CPU cores always give exactly four times the performance of one core.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.